



Amr Ahmed Abdelfattah Ahmed WalyEldeen



Personal information

Family name : WalyEldeen.
 First name : Amr.
 Date of Birth : 10.07.1995.
 Nationality : Egyptian.
 Marital Status : Married (1 Child)
 Position Title : Assistant lecturer of Cell, Histology, & Genetics.
 Present Address : Cellular and Molecular Biosciences Research Lab, Department of Zoology,
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All academic profiles and professional information—including Google Scholar, Scopus, ORCID iD, LinkedIn, and Cairo Scholar—are available on my website.

Personal Academic Website: <https://amrwalyeldeen.github.io/>

Scan the QR code to access all links:



Summary

Assistant Lecturer and Researcher in Cell Biology, Histology, and Genetics at Cairo University, with experience in cancer cell biology, molecular oncology, and extracellular vesicle biology. My research focuses on understanding the molecular mechanisms underlying cancer progression, metastasis, and therapeutic resistance through advanced proteomic and bioinformatic approaches. I am pursuing Ph.D. that involves proteomic profiling of circulating exosomes from breast cancer patients to identify proteome signatures of exosomal cargo, aiming to develop cost-effective, non-invasive liquid biopsy tools for early breast cancer detection.

Language proficiency

Arabic : Fluent spoken and written (mother tongue).
 English : Fluent spoken and written.

Academic degrees

Year	Degree	Institution	Country	Field / Subjects	Grade
2023	PhD Courses	Cairo University	Egypt	Cell Biology, Histology, and Genetics	Excellent, 94.625%
2021	Master's Degree	Cairo University	Egypt	Cell Biology, Histology, and Genetics	N/A
2018	Pre-Master Courses	Cairo University	Egypt	Cell Biology, Histology, and Genetics	Very Good, 84%
2017	B.Sc.	Cairo University	Egypt	Zoology and Chemistry	Excellent with Honors, GPA 4.2/5, 92%

Professional background

From	To	Position	Institution	City, Country
09.2022	Present	Assistant Lecturer	Zoology Department, Faculty of Science, Cairo University	Giza, Egypt
12.2018	09.2022	Teaching Assistant	Zoology Department, Faculty of Science, Cairo University	Giza, Egypt
10.2017	10.2018	Chemist	Alfa Laboratories	Giza, Egypt



Awards

Award Name	Awarding Institution	Year
Cairo University International Publication Award	Cairo University	2023, 2024, 2025
Appreciation Award in the Faculty of Science Street Photography Competition	Faculty of Science	2018
Faculty of Science Award for Top Graduating Students	Cairo University	2018
Department of Zoology Award for the Top-Ranked Student (1st place)	Cairo University	2018
Dr. Zain Al-Abidin Award for the First and Second-Ranked Students of the Faculty of Science Graduating Class	Cairo University	2018
Syndicate of Scientific Professions Award for Top Graduates of the Faculty of Science	Syndicate of Scientific Professions	2017

Projects

1. Extracellular vesicle-derived proteoglycans as non-invasive liquid biopsy-based cancer biomarkers: Towards novel breast cancer screening (project No. 46282)

- Duration: 2 years
- Role: member of the research team (Researcher B: M.Sc. holder)
- Funded by Science, Technology & Innovation Funding Authority (STDF)
- Budget: 1,991,170 LE

Publications: The project has resulted in Three research articles two published and one under review.

2. Proteomic profiling of exosomes: Discovery of novel diagnostic biomarkers for lymph node metastasis in breast cancer (project No. Respect1-10052)

- Duration: 2 years
- Role: member of the research team (Researcher B: M.Sc. holder)
- Funded by The Academy of Scientific Research & Technology (ASRT)
- Budget: 1,490,000 LE

Publications: The project has resulted in one published research article.

PhD thesis

Thesis title: Proteomic Profiling of Circulating Exosomes in Breast Cancer Patients.

Thesis supervisors: Prof. Dr. Sherif Abdelaziz Ibrahim (Professor of Cancer Cell Biology, Zoology Department, Faculty of Science, Cairo University), Dr. Ghada Mohamed Abdel-Salam (Associate Professor of Pathology National Cancer Institute, Cairo University), Dr. Sherif Nasser Mohamed Taha (Lecturer of Surgical Oncology, National Cancer Institute, Cairo University).

Thesis Publications:

Published articles:

1. **WalyEldeen, A. A.,** Mohamed, G., Taha, S. N., Gameel, A. M., Talat, L. Y., Hassan, H., & Ibrahim, S. A. (2026). Plasma small-extracellular vesicles' proteomic signature in neoadjuvant chemotherapy-naïve breast cancer patients. *PloS one*, 21(5), e0348500.

Under review:

2. **Title:** Identification of cathepsin L and urokinase-type plasminogen activator in plasma-derived small extracellular vesicles of breast cancer patients



Authored by: Amr Ahmed WalyEldeen, Ghada Mohamed, Sherif Nasser Taha, Abdallah M. Gameel, Liali Yousef Talat, Mostafa I. Abdelglil, Sherif Abdelaziz Ibrahim, and Hebatallah Hassan.

Journal: Biochimica et Biophysica Acta (BBA) - General Subjects Journal

Status: Under review (**Manuscript ID:** BBAGEN-26-265).

Related publication to the project funding the thesis:

3. Talat, L. Y., **WalyEldeen, A. A.**, Mohamed, G., Ibraheem, M. H., Maher, M. M., Ibrahim, S. A., ... & Götte, M. (2026). Large Extracellular Vesicle-Derived Latent MMP-8 and Gelatinolytically Active MMP-2 as Potential Circulating Markers for Lymph Node Metastasis in Breast Cancer. *Cancers*, 18(9), 1464.

Thesis Titles

- **Master's thesis**

Evaluation of the Potential Anticancer Effect of a Novel Chalcone Derivative in Ehrlich Solid Carcinoma-Bearing Mice. (Department of Zoology, Faculty of Science, Cairo University).

Thesis supervisors: Dr. Sherif Abdelaziz Ibrahim (Professor of Cancer Cell Biology, Zoology Department, Faculty of Science, Cairo University). Prof. Dr. Salwa Sabet (Professor of Molecular Biology, Zoology Department, Faculty of Science, Cairo University). Prof. Dr. Haidan Mosatafa (Professor of Molecular Biology, Zoology Department, Faculty of Science, Cairo University).

The thesis was examined and reviewed by

Prof. Dr. Burkhard Greve (Department of Radiotherapy-Radio oncology- University Hospital Munster Email: Burkhard.greve@ukmuenster.de) with overall assessment: Acceptance in the thesis present form and degree awarded.

- **Prof. Dr. Ahmed Said Alazzouni** (Department of Histology and Histochemistry, Faculty of Science, Helwan University. Email: drahmedalazzouni@gmail.com) with overall assessment: Acceptance in the thesis present form and degree awarded.

- **Thesis Publications:**

1. **WalyEldeen, A. A.**, El-Shorbagy, H. M., Hassaneen, H. M., Abdelhamid, I. A., Sabet, S., & Ibrahim, S. A. (2022). [1, 2, 4] Triazolo [3, 4-a] isoquinoline chalcone derivative exhibits anticancer activity via induction of oxidative stress, DNA damage, and apoptosis in Ehrlich solid carcinoma-bearing mice. *Naunyn-schmiedeberg's Archives of Pharmacology*, 395(10), 1225-1238.
2. **WalyEldeen, A. A.**, Sabet, S., El-Shorbagy, H. M., Abdelhamid, I. A., & Ibrahim, S. A. (2023). Chalcones: Promising therapeutic agents targeting key players and signaling pathways regulating the hallmarks of cancer. *Chemico-biological interactions*, 369, 110297.

Training and workshops

Year	Institution
2026	Metagenomics, Microbiome & 16s rRNA sequencing workshop organized by Colors Educational Hub, Cairo, Egypt
2026	Introduction to single cell genomics workshop, organized by EG-CompBio, Faculty of Science, Cairo University
2025	How to Create Graphical Abstracts That Boost the Impact of Medical Research online Event by Editage.
2025	Cairo University's Faculty and Leadership Development Centre held a training course on the application of artificial intelligence.
2025	Computational Biophysics Workshop organized by the Department of Biophysics, Faculty of Science, Cairo University (3-21 August 2025)
2024	Metabolomics day was held at the 57357 Children's Hospital.



2024	Cairo University's Faculty and Leadership Development Centre held a training course on the effective presentation skills
2023	Biostatistics workshop in Assisted Reproduction Technologies, organized by the Egyptian Society of Embryology and Assisted Reproduction Technologies.
2022	Workshop on the role of genetic testing in Assisted Reproduction Technologies, organized by the Egyptian Society of Embryology and Assisted Reproduction Technologies.
2021	Workshop on the role of the andrology laboratory in Assisted Reproduction Technologies, organized by the Egyptian Society of Embryology and Assisted Reproduction Technologies.
2020	I attended an online seminar, "Targeting metastatic breast cancer" by Denis Wirtz, Johns Hopkins University.
2020	A training course on using technology in education was held at Cairo University's Faculty and Leadership Development Centre.
2020	A training course in "International publishing of scientific papers" was held at Cairo University's Faculty and Leadership Development Centre.
2019	Participant in the international winter school on bioinformatics held at Nile University in collaboration with Rostock University.
2019	Participant in the international summer school on bioinformatics held at Nile University in collaboration with Rostock University.
2019	Biostatistics program (8 levels) held at the Egyptian Association for Animal Research Advancement, Faculty of Science, Cairo University.
2019	Workshop of Husbandry, handling, and basic techniques in laboratory animals based on FELASA criteria, organized by EAARA in collaboration with Utrecht university
2019	Attended the Symposium "How to write a successful animal care protocol" held at the Faculty of Science at Cairo University.
2019	Training course in critical thinking held at the Faculty and Leadership Development Centre at Cairo University.
2019	Internship program at Aswan Heart Center (Magdi Yacoub Foundation for cardiac research and disease) in wet-lab techniques at the Life Science department.

Conference participants

- Science at the Core of Innovation**, Cairo, Egypt, April 7–8, 2026.
Poster Presentation: Amr Ahmed WalyEldeen, Ghada Mohamed, Sherif Nasser Taha, Abdallah M. Gameel, Liali Yousef Talat, Hebatallah Hassan, Sherif Abdelaziz Ibrahim. *Plasma Small-Extracellular Vesicle Proteomic Signature in Neoadjuvant Chemotherapy-Naïve Obese Breast Cancer Patients*.
Award: Winner of 2nd Place for Best Research Idea.
- ASTRO 2025: 67th Annual Meeting**
Abstract: Troschel F, Brücksken KA, Meier FK, Meißner C, Poschmann G, Korsching E, Hassan H, WalyEldeen AA, Ibrahim SA, Samer RM, Sicking M. *Knockdown and Inhibition of RNA-Binding Proteins Musashi-1 and Musashi-2 Modulate DNA Repair and Radiosensitize Triple-Negative Breast Cancer*. International Journal of Radiation Oncology, Biology, Physics. 2025 Sep 1;123(1):e515.
- Pathology Insights Conference**, Cairo, Egypt, July 9, 2025.
 Attendee.
- International Conference of the Society of Endometriosis in Egypt — Endo Egypt**, Cairo, Egypt, February 22, 2025.
 Attendee.
- First Annual "Next Generation Cancer Research and Diagnosis" Conference**, Cairo, Egypt, 2014.
 Attendee.



Organization Experience

Year	Institution
2026	Organizing the Zoology Department Student Activities Exhibition held at Faculty of Science, Cairo University on 9 February 2026
2022	Presenter and organizer of Hands on basic and advanced cell culture techniques the Egyptian Society of Embryology and Assisted Reproduction Technologies.
2020	Presenter of online workshop “Introduction to Bioinformatics Tools”.
2019	Organizer of the workshop “Sustainable development, scientific research, and labor market” held at the Faculty of Science, Cairo University.
2018	Organizer of workshop “Requirements of Works and different aspects for working for graduates of Zoology Department”.

Skills

Laboratory Skills

DNA & RNA extraction
 PCR and Real-Time PCR
 Cell culture and sub-culturing of cell lines
 miRNA extraction
 Western blot
 Immunohistochemistry
 Comet assay
 Animal handling
 ELISA
 3D organoid culture and maintenance
 Flow cytometry

Computational & Analysis

Protein modeling
 Molecular docking
 Molecular dynamics simulations
 Meta-data analysis
 Bioinformatics workflows

Technical & Digital

Program development using AI tools
 Website creation using AI-assisted workflows

Society Memberships

Member, Egyptian Society of Embryology and Assisted Reproduction Technologies.

Leadership & Academic Service

Committee Rapporteur, Student Activity Committee, Zoology Department, Faculty of Science, Cairo University.

Teaching experiences

Weekly teaching labs:

Practical General Zoology for first-year students; Practical Cell Biology and Practical Genetics for second-year students; and Practical Histochemistry and Practical Molecular Biology for fourth-year students.



List of publications

2026

1. **WalyEldeen AA**, Mohamed G, Taha SN, Gameel AM, Talat LY, et al. (2026) Plasma small-extracellular vesicles' proteomic signature in neoadjuvant chemotherapy-naïve breast cancer patients. *PLOS ONE* 21(5): e0348500. <https://doi.org/10.1371/journal.pone.0348500>.
2. Talat, L.Y.; **WalyEldeen, A.A.**; Mohamed, G.; Ibraheem, M.H.; Maher, M.M.; Ibrahim, S.A.; Hassan, H.; Götte, M. Large Extracellular Vesicle-Derived Latent MMP-8 and Gelatinolytically Active MMP-2 as Potential Circulating Markers for Lymph Node Metastasis in Breast Cancer. *Cancers* 2026, 18, 1464. <https://doi.org/10.3390/cancers18091464>
3. Ibrahim, N.S., WalyEldeen, A.A., Ibrahim, S.A. et al. Synthesis and cytotoxicity of novel cyanochalcones: induction of cell cycle arrest, apoptosis, and autophagy in HEP2 and MCF-7 cells. *Naunyn-Schmiedeberg's Arch Pharmacol* 399, 8867–8882 (2026). <https://doi.org/10.1007/s00210-025-04940-z>.
4. Zhran, E. A., Aboushousha, T., **WalyEldeen, A. A.**, Hegab, F., & Ibrahim, S. A. (2026). Cullin1 overexpression is associated with colorectal cancer progression and poor prognosis. *Tissue and Cell*, 98, 103194. <https://doi.org/10.1016/J.TICE.2025.103194>

2025

5. Troschel, F., Brücksken, K. A., Meier, F. K., Meißner, C., Poschmann, G., Korsching, E., Hassan, H., **WalyEldeen, A. A.**, Ibrahim, S. A., Samer, R. M., Sicking, M., Götte, M., Greve, B., & Eich, H. T. T. (2025). Knockdown and Inhibition of RNA-Binding Proteins Musashi-1 and Musashi-2 Modulate DNA Repair and Radiosensitize Triple-Negative Breast Cancer. *International Journal of Radiation Oncology*Biophysics*, 123(1), e515. <https://doi.org/10.1016/j.ijrobp.2025.06.2677>
6. Talat, L. Y., Mohamed, G., Ibraheem, M. H., **WalyEldeen, A. A.**, Hassan, H., & Ibrahim, S. A. (2025). SDC2 and FN as cargo proteins in circulating extracellular vesicles in obese breast cancer patients with lymph node metastasis. *Scientific Reports* 2025 15:1, 15(1), 32498-. <https://doi.org/10.1038/s41598-025-17638-2>
7. Sayed, A. A., Abdullah, M. S., **WalyEldeen, A. A.**, Sayed, R. M. S., Gabre, R. M., Ibrahim, S. A., & Hassan, H. (2025). Cicer arietinum extract as antitumor and protective agent against Ehrlich Solid Carcinoma-bearing mice. *BMC Complementary Medicine and Therapies* 2025 25:1, 25(1), 325-. <https://doi.org/10.1186/S12906-025-05061-Z>
8. Kotb, M. A., Abdelmawgood, I. A., **WalyEldeen, A. A.**, Abdelaziz, M. H., Khalaf, M. L., Mohamed, A. S., Dakrory, A. E., & Ibrahim, S. A. (2025). Chrysin-loaded PLGA nanoparticles alleviate the implantation of endometriotic lesions via attenuation of peritoneal inflammation and downregulating NF-κB activation-driven expression of angiogenic factors. *Journal of Drug Delivery Science and Technology*, 109, 106984. <https://doi.org/10.1016/J.JDDST.2025.106984>
9. Hassanein, H. H., **WalyEldeen, A. A.**, Sayed, R. M. S., Taha, A. A. A., Ibrahim, S. A., & Hassan, H. (2025). Glypican3 and serglycin as potential biomarkers involved in the pathogenesis of ovarian endometriosis. *Tissue and Cell*, 95, 102867. <https://doi.org/10.1016/J.TICE.2025.102867>
10. Mahmoud, M., **WalyEldeen, A. A.**, Samie, M. A., Sadek, A. M., Elakhras, S., Samir, M., Ahmed, O., Sayed, R. M. S., Baiomy, A. A., Ibrahim, S. A., & Hassan, H. (2025). Overexpression of miR-21-5p as a potential pathogenesis marker for ovarian endometriosis. *The Journal of Basic and Applied Zoology* 2025 86:1, 86(1), 1-. <https://doi.org/10.1186/S41936-024-00414-5>

2024

11. **WalyEldeen, A. A.**, Sabet, S., Anis, S. E., Stein, T., & Ibrahim, A. M. (2024). FBLN2 is associated with basal cell markers Krt14 and ITGB1 in mouse mammary epithelial cells and has a preferential expression in molecular subtypes of human breast cancer. *Breast Cancer Research and Treatment* 2024 208:3, 208(3), 673–686. <https://doi.org/10.1007/S10549-024-07447-Y>

2023



12. **WalyEldeen, A. A.**, Sabet, S., El-Shorbagy, H. M., Abdelhamid, I. A., & Ibrahim, S. A. (2023). Chalcones: Promising therapeutic agents targeting key players and signaling pathways regulating the hallmarks of cancer. *Chemico-Biological Interactions*, 369, 110297. <https://doi.org/10.1016/J.CBI.2022.110297>
13. Emad, B., **WalyEldeen, A. A.**, Hassan, H., Sharaky, M., Abdelhamid, I. A., Ibrahim, S. A., & Mohamed, H. R. (2023). Yttrium Oxide nanoparticles induce cytotoxicity, genotoxicity, apoptosis, and ferroptosis in the human triple-negative breast cancer MDA-MB-231 cells. *BMC Cancer* 2023 23:1, 23(1), 1151-. <https://doi.org/10.1186/S12885-023-11649-W>

2022

14. **WalyEldeen, A. A.**, El-Shorbagy, H. M., Hassaneen, H. M., Abdelhamid, I. A., Sabet, S., & Ibrahim, S. A. (2022). [1,2,4] Triazolo [3,4-a]isoquinoline chalcone derivative exhibits anticancer activity via induction of oxidative stress, DNA damage, and apoptosis in Ehrlich solid carcinoma-bearing mice. *Naunyn-Schmiedeberg's Archives of Pharmacology* 2022 395:10, 395(10), 1225–1238. <https://doi.org/10.1007/S00210-022-02269-5>
15. **Alghamdi, A. A. A.**, A. WalyEldeen, and S. Ibrahim, "Nanoparticles as a Therapeutic Approach for Tumor Angiogenesis", *Innovative Approaches for Nanobiotechnology in Healthcare Systems*: IGI Global, 2022.

Hobbies

Photography: Participant at the second, third, and fourth annual gallery for photography in Sakia Elsayy, Participant at the second competition on street photography at Cairo University, and winner of the appreciation award

References

Professor Dr. **Sherif Ibrahim**, Zoology Department, Faculty of Science, Cairo University Email: isherif@cu.edu.eg

Professor Dr. **Salwa Farouk Sabet**, Zoology Department, Faculty of Science, Cairo University Email: salwasabet@cu.edu.eg

Associate Professor Dr. **Ayman Maher Ibrahim**, Zoology Department, Faculty of Science, Cairo University Email: sci_ayman@cu.edu.eg